GRAVITON.TOOLS · RELEASE NOTES

Manifold 0.9.0

Version 0.9.0 (build 21) · 29 October 2026 · Beta for testers

> **Before you upgrade: once you have opened 0.9.0, don't go back to 0.8.4.** 0.9.0 moves your saved streams into the macOS Keychain. 0.8.4 can't connect to them after that, and editing or deleting a saved stream in 0.8.4 can lose it for good. Your license carries over in both directions; your saved streams do not.

Summary

0.9.0 is the feature-complete beta before Manifold 1.0. Live streams now hold sound and picture in sync over long sessions, with a new A/V menu to measure and save a stream's offset. Color handling is rebuilt: every window can show the picture as macOS does, as a reference monitor would, or untouched, and you can override any file's or stream's color tags or apply a LUT. The DeckLink output now tells your monitor when the picture is HDR, fills its output mode whatever the picture size, plays interlaced files out as 1080i and plays files back smoothly. Settings and keyboard shortcuts have been reorganised, and your license and saved streams now live in the macOS Keychain.

Highlights

- **Sound and picture stay in sync on live streams.** SRT, WHEP and NDI were each out by a fixed or drifting amount in 0.8.4. They now hold sync over sessions of several hours.
- **Audio sync calibration and per-stream offsets**, in a new A/V menu: measure a stream's offset with a Manifold sync clip, nudge it by the millisecond, and save it with the stream.
- **macOS, Reference or Bypass, per window:** see the picture as macOS shows it, with the BT.1886 curve a reference monitor uses, or with no color management at all, and read exactly what happens between the source and your screen.
- **Live streams show the color they declare,** including PQ and HLG HDR over SRT and WHEP, and HEVC now plays over SRT, including 10-bit HDR.
- **SDI and HDMI output that follows the picture:** HDR signalling, any picture size, 1080i for interlaced files, and smooth file playback.

New

COLOR

- **Display transform, per window.** OS lets macOS color-manage the picture for your display. Bypass sends the picture's values to the screen untouched. Choose it from the control bar or the Color menu. A window in Bypass says so in its title.
- **Reference display transform, per window:** the picture shown with the BT.1886 curve a reference monitor uses (gamma 2.4 on a display with a true black). In the transform chain readout you can tell Manifold, per display, that the display is calibrated to Rec. 709 / BT.1886, in which case Reference sends the picture's values untouched, or enter the display's white and black so Reference follows BT.1886 for it. These settings apply in Reference only. Scopes, SDI output and exported frames are not affected by the display transform.
- **Hold to compare.** Hold C to see the other of macOS and Reference while the key is down, or Shift-C to see Bypass. Let go and the window is back where it was. With a LUT on, hold Option-C to see the picture without it. The window title and an amber badge say what you are comparing against.
- **The display chain readout** shows the source's color and whether it was declared or assumed, the transform in use, your display and its curve, and, for SRT and WHEP, how much the stream is buffered.
- **HDR headroom in the readout:** "display is in SDR mode" when macOS HDR is off for that display, or how much headroom it offers. It follows the window to another display and updates when HDR is switched or the brightness changes. It only reports: Manifold does not cap or tone-map the picture.
- **Tag mismatch warning.** The readout warns when an H.264 or HEVC file's container color tags disagree with what its own video stream declares. The container's tags are still used.
- **Color overrides on files and every stream type.** Pick Rec. 709, Rec. 2020 PQ, Rec. 2020 HLG, P3-D65 PQ or Rec. 2020 SDR from the Color control, or choose Color ▸ Next Tag Override. On a file the override lasts until you open another file. On SRT, WHEP and HLS streams (as on NDI) you can save it with the saved stream, and it comes back every time you connect. The picture, scopes, SDI color tag and readout change together, and the window title says so.
- **Override the tags one by one:** primaries, transfer and matrix separately, in the Color control's Override section. Combinations Manifold can't display as stated are greyed out with the reason.
- **Reset to File's Tags** (or Reset to Stream's Tags) at the top of the Color control undoes an override at once.
- **LUTs.** Load LUT… in the Color control applies a .cube LUT (1D, 3D, or shaper + 3D) to the window's picture, and the scopes, SDI output and exported frames see it too. It stays on when you open another source. Interpolation is tetrahedral by default, or trilinear. LUT Outputs says what the LUT produces: leave it at Same as Source unless the LUT converts to another color space, such as Rec. 709 to PQ. Recent LUTs remembers each LUT with its output space, and Bypass LUT turns a LUT off without unloading it. The Color control, window title and readout always say when a LUT is on or bypassed. A file that isn't a valid .cube is refused with a message saying why. Results were checked against DaVinci Resolve.

SDI/HDMI OUTPUT

- **HDR signalling.** Each frame tells the card its color space and transfer (SDR, PQ or HLG), so a reference monitor that follows the signal can switch on its own. PQ also sends HDR10 metadata over HDMI: the source's own values, or standard defaults when the source has none, and the DeckLink menu says which. HLG sends the transfer only.
- **SDI scaling** (Settings ▸ DeckLink Output): Fit to mode, the default, scales a picture that isn't an output mode's size to fill the mode, keeping its shape, with black around it. 1:1 pixels sends it at its own size instead.
- **SDI output modes** (Settings ▸ DeckLink Output): HD/UHD only, the default, or Include DCI to let the output follow a source into 2K DCI (2048×1080) and 4K DCI (4096×2160).
- **More frame rates.** The output follows sources at 47.95, 48, 95.90, 96, 100, 119.88 and 120 fps, in 1080p, 2160p and DCI, and 720p50, 59.94 and 60. Before, those rates went out at the nearest of the standard eight.
- **Interlaced files go out interlaced.** A 1080i file at 25, 29.97 or 30 frames a second goes out as 1080i50, 1080i59.94 or 1080i60, with both fields exactly as they are in the file. A bottom-field-first file is sent as it is, and the DeckLink menu warns that its fields will play out of order on the top-field-first 1080i signal.

LIVE STREAMS

- **Audio offset for live streams**, in the A/V menu: ⌥] and ⌥[move the sound 1 ms later or earlier (add ⇧ for 10 ms). Save the offset with a saved stream and it comes back every time. A badge and the window title show when an offset is on.
- **Calibration** (A/V ▸ Calibrate…): loop a Manifold sync clip on the sender and Manifold measures how far apart sound and picture are, then offers the correction. It never applies anything by itself. A/V ▸ Save Sync Clip… saves the clip that matches the stream's frame rate, and Help ▸ Download Sync Clips… gets the full set.
- **HEVC over SRT,** including 10-bit HDR.

FILES AND INTERLACED

- **Scan row in the Inspector:** whether a file is progressive or interlaced, which field comes first, and where that came from. If the container disagrees with the picture data, the row says so. Frame Rate gives the field rate too, for example "29.970 fps (59.94 fields/s)".
- **Field views for interlaced files.** Color ▸ Fields, or the Fields pulldown in the control bar, offers Weave (both fields as stored, the default), Field 1, Field 2 and Deinterlace. These are for the window only: scopes, SDI and HDMI output and exported frames always show the file as stored. While a field view is on, a "Display only" badge, the window title and a marker in the picture's corner say so.

INTERFACE AND SHORTCUTS

- **Keyboard Shortcuts…** (in the Manifold menu and in Help) lists every shortcut with the keys your keyboard layout prints. Search by name, or press a key to see what it does. Click a shortcut to change it: Manifold warns before taking a key from another action, refuses keys macOS uses, and can reset one shortcut, a group, or all of them.
- **Highlight color** (Settings ▸ General): the color Manifold uses for the playhead, controls that are on, on-picture overlays and override text. Yellow by default. Green still means live.

- **Line playhead:** the timeline's playhead is now a line with a grab tab, centred exactly on the current frame. Prefer the round handle? Settings ▸ Playback ▸ Playhead ▸ Ball. The line can follow the highlight color or have its own.
- **Transport keys:** hold K and press J or L to step one frame, and keep holding to play at half speed. Home and End (or Option-Left and Option-Right) go to the first and last frame. ⌘L turns looping on and off. ⌘F enters and leaves full screen, as do Control-⌘-F and fn-F.
- **Volume keys:** ⌘↑ and ⌘↓ change the volume by a tenth, and ⌥⌘↓ mutes and unmutes (View ▸ Audio).
- **Output and source keys:** ⇧⌘D starts and stops the DeckLink output, ⌥⌘⌫ disconnects the live source, and ⌥⌘4 shows and hides the transform chain.

LICENSING AND PRIVACY

- **Activate without an internet connection.** Manifold works straight away and finishes registering your Mac the next time it can reach the license server.

Improved

COLOR

- The display chain readout marks a color override, and a LUT that is on, in the same amber as the Color control.
- Color menu wording: "Next Color Interpretation" is now "Next Tag Override", "Reset Color Interpretation to Auto" is now "Reset Tags to Source", and the Color control's "Custom" section is now "Override". Saved streams and shortcuts load as before.

SDI/HDMI OUTPUT

- While DeckLink output is running, your Mac and its displays don't go to sleep, so a long playout isn't interrupted. They can sleep again as soon as output stops.
- When you stop the output, close the window that owns it, or quit, Manifold sends a final black frame marked as SDR before letting go of the card, so the card and the display after it are left on black in SDR rather than on the last picture, or in HDR.

LIVE STREAMS

- Lower latency: WHEP plays 100 ms sooner and NDI 50 ms sooner, with sound and picture still in sync.
- SRT and WHEP show the color the stream declares. A stream tagged HDR (PQ or HLG) is shown and scoped as HDR. A stream that declares nothing is still treated as Rec. 709.
- SRT streams show picture much sooner. A static slate or bars feed at a low bitrate used to take a minute or more to appear; it now takes a second or two.
- WHEP from Cloudflare Stream keeps sound and picture in sync for long sessions, including when sent out over SDI.
- SRT buffering adapts to the sender. Encoders that send audio in large chunks, or use B-frames, get a slightly larger buffer automatically, and the chain readout says how much and why.
- SDI output follows an SRT stream's real frame rate, even when the stream labels it wrongly.

- SRT accepts HEVC and H.264 4:2:2 and 4:4:4 at full color resolution. A stream this Mac can't decode is refused with a message saying so, never quietly converted.
- Finer color detail in 8-bit H.264 streams over SRT and WHEP, which is what OBS and most encoders send.

FILES AND INTERLACED

- 4:2:2 and 4:4:4 files (ProRes, DNxHD/DNxHR, H.264/HEVC 4:2:2) keep their full color resolution through playback, scopes, SDI and export.
- ProRes MXF decodes with Apple's hardware decoder: faster, and identical to the same ProRes in .mov.
- Finer color detail in 8-bit H.264 and HEVC files: color edges between lines are as sharp as the file has them, in the window, scopes, exported frames and SDI.
- Interlaced files with 4:2:0 color (H.264, MPEG-2) no longer mix the two fields' color. Interlaced ProRes and DNxHD were already right. (See Known limitations for what macOS's decoder still blends.)

INTERFACE AND SHORTCUTS

- Settings has tabs (General, Files, Playback, Scopes & Guides, DeckLink Output, Sources, License) and can be resized. Manifold ▸ License… opens the License tab. Your settings are unchanged.
- Shortcuts are in the menus now, where you can see them, and some keys changed: Inspector ⌘I (was I), Always Show Controls ⌥⌘T (was Tab), Show/Hide Scopes ⌥⌘S, Export Frame ⌥⌘E, and the display transforms ⌥⌘1, ⌥⌘2 and ⌥⌘3. Every menu item shows its key.
- Several Control-Option keys became menu items: Next Tag Override is in the Color menu (no key unless you set one), Disconnect Live Source is ⌥⌘⌫, Start/Stop DeckLink Output is ⇧⌘D, and each scope's settings are in its own menu.
- **Export Diagnostics… has moved to the Help menu,** next to Export Frame for Support….
- Export Frame writes the image only. **Help ▸ Export Frame for Support…** saves the current frame with the measurement files support may ask for. It asks where to save them, and the files include the frame's picture.
- The guides popover is laid out as Broadcast safe zones first, then one "Frame guide" choice (None, formerly Off; the aspect ratios and Custom; the social ratios), then Social safe zones. Settings ▸ Scopes & Guides uses the same sections. Saved guide choices load as before.
- One yellow for "on": the pin when pinned, Loop when on (it was green), an override or LUT, and the hold-to-compare badge all use the highlight color. Green now means live only: DeckLink output running, or a connected stream.
- The volume control no longer shakes when you hover or drag it.

LICENSING AND PRIVACY

- Your license email and machine ID are kept only in the macOS Keychain, no longer in Manifold's preferences file.
- Saved streams are kept in the macOS Keychain in full: the address, with any stream key or token in it, and the passphrase. The preferences file keeps only each stream's name and server.
- When you edit a saved stream, its address stays hidden unless you click Show saved URL. Leave the field blank to keep the saved address.

- Licensing works as it does in every other Graviton app. Manifold checks your license with the server at most once a week, and never locks you out for being offline.
- A new activation identifies your Mac by its hardware, so reinstalling Manifold or macOS doesn't use up another activation. Macs already activated keep the activation they have.
- Activation messages say how many of your license's activations are in use, and a license that can no longer be used says so instead of showing the trial screen.
- Diagnostics reports no longer include any part of your license email or key, and leave out IP addresses, folder paths, computer and NDI sender names and display serial numbers. File names are left out unless you tick "Include file names".
- Manifold no longer keeps a disk cache of its license, update and stream requests, and removes the one earlier versions left behind. Its log no longer records network addresses or SRT stream IDs.
- NDI looks for sources on your network only when you open the NDI source list or connect to a source.

PERFORMANCE

- Scopes use far less of the processor while playing: each scope redraws only its own trace.
- Opening a file in one window no longer makes every other window recompute its scopes.

Fixed

COLOR

- P3-D65 and DCI-P3 sources went out over SDI labelled Rec. 2020 without being converted, so they looked oversaturated on a monitor following the label. They are now converted for SDI.
- Sources using the Rec. 601 color matrix were shown about 14 % too saturated, in the window, scopes, SDI and exported frames.
- Streams using the BT.470BG matrix, common from European encoders, had a small color shift.
- Files tagged Rec. 601 or BT.470BG were measured by the scopes and sent over SDI as Rec. 709.
- Rec. 2020 SDR, DCI-P3, SMPTE-C and EBU sources were shown with Rec. 709 primaries, so they looked less saturated than they are.
- MXF and DNx files whose container and video stream declare different color matrices used one for the picture and the other for the scopes and SDI. The container's color is now used for all of them.
- DNxHR 444 files in a .mov could look slightly flat, with lifted blacks and weaker color.
- 8-bit sources read slightly high on the scopes (mid-grey 514, white 943); they now map exactly.
- HLS streams in P3 were shown with Rec. 709 colors while the scopes said P3.
- The scopes could keep an earlier source's color settings after switching streams or after a stream ended. They now always describe what is playing.

SDI/HDMI OUTPUT

- While a file played out over SDI or HDMI, the output now and then showed one frame twice and then skipped the next. File playback out of the card is now smooth.
- SDI audio dropped out every few seconds during file playback.

- SDI and HDMI were black for any source whose size isn't an output mode, such as 720p at 23.976, vertical or square video. The picture now fills the closest mode, and the DeckLink menu says how it was placed or scaled.
- A live stream that changed size mid-session, such as HLS stepping between renditions, could restart the output at a new mode. The mode is now chosen once when the stream starts.
- When a stream's color changed during output, the output kept the old color label, and it could go back to an earlier source's label when a file with sound was opened.
- An HDR10 file's first frames could carry default HDR metadata instead of the file's own.
- Starting output while paused sent black until you pressed Play.
- When another application, such as DaVinci Resolve, was using the DeckLink output, starting output failed with no explanation. Manifold now says the output is in use by another application.

LIVE STREAMS

- SRT sound played about 200 ms behind the picture, and NDI sound about a quarter of a second behind.
- WHEP sound and picture were out by a different amount every session, and drifted further apart over long sessions.
- SRT sound from Cloudflare could sound distorted; SRT streams from some encoders had no sound at all; and SRT sound broke up on low-bitrate streams.
- After reconnecting, an SRT stream could stop about 17 seconds in with "The stream stopped sending video".
- After an NDI or HLS stream ended, a file played next ran its picture ahead of its sound.
- H.264 4:2:2 streams over SRT were shown at 4:2:0 without saying so, and all-intra HEVC 4:2:0 streams were refused.
- The stream button turned green for NDI only. It now turns green for any connected stream.
- During and after a stream, the window's name in the Window menu and Mission Control still showed the file opened before it.
- The NDI source list no longer jitters or reshuffles while it is open; a new source is added at the end.

FILES AND INTERLACED

- Stepping frame by frame through 23.976, 29.97 and 59.94 fps files showed the previous frame, and the last frame couldn't be reached.
- A frame step could occasionally land one to three frames past its target, and tapping ← or → quickly could skip steps. Every press now moves exactly one frame.
- On MXF and DNx files, a typed timecode showed the frame after the one typed, and releasing the scrubber could land one frame late.
- Some camera files rewrapped by older tools (for example ARRI ProRes clips) couldn't be opened because of a container error. Manifold now reads around the error, says so, and leaves the file unchanged.
- Some files with PCM audio played silently because of a malformed channel layout, and one such file could make Manifold quit. They now play with their audio.
- When a file's audio can't be read, the meters now say "Audio not read" and the Inspector says why.
- Closing a window while its file was still opening could leave the sound playing with no window.

- iPhone clips with rotation could open mirrored.
- Vertical videos could open taller than a laptop screen and couldn't be resized smaller.

INTERFACE AND SHORTCUTS

- ⌥⌘S (View ▸ Show/Hide Scopes) could stop working after the first time you used a menu.
- In light appearance the player window stays dark, as QuickTime's does. Settings, Keyboard Shortcuts and About follow your Mac's appearance.
- File ▸ Close and Window ▸ Minimize were greyed out whenever the controls had faded, so ⌘W and ⌘M only worked with the pointer over the window.

PERFORMANCE

- A paused window with the scopes open kept using about a quarter of the processor, so with several windows open Manifold became slow to respond. A paused window now uses almost none.
- Closed windows kept their memory until Manifold quit. They are now released when the window closes.

Known limitations

COLOR

- PQ or HLG on primaries other than Rec. 2020 or P3-D65 can't be shown as stated, and those overrides are greyed out.

SDI/HDMI OUTPUT

- A file's picture and sound reach SDI and HDMI about 11 ms later than in 0.8.4, the cost of the smooth playback fix. They stay together on the card; if you line SDI up against another source, check the offset again.
- Over HDMI, an HLG picture goes out without its HLG signal, so the display isn't told to switch to HLG. SDI carries HLG.
- SDI audio goes silent for up to about a tenth of a second at each seek, scrub release and loop point during file playback.
- Files at 95.90 fps and above don't yet play at full speed, so SDI repeats frames at those rates. On a DeckLink 8K Pro, 2160p above 60 fps doesn't run cleanly yet.

LIVE STREAMS

- Interlaced streams (SRT, WHEP, HLS, NDI) still go out over SDI as progressive, with both fields woven into each frame. An interlaced NDI source arrives already deinterlaced by NDI.
- After a network stall of a second or more, live sound can stay 10–20 ms late for up to a minute.
- OBS stamps 23.976 fps WHIP video very slightly fast. Through MediaMTX on its default settings, sound and picture drift apart by about a quarter of a second an hour over WHEP. Set `useAbsoluteTimestamp: true` on the MediaMTX path to fix it. Cloudflare Stream corrects it on its own.

FILES AND INTERLACED

- Interlaced files that aren't 1080 lines, and 1080 interlaced at 23.98 or 24, have no interlaced output mode. They go out progressive at their own size, and the DeckLink menu says so.
- On 8-bit H.264 files macOS's decoder still blends some color between neighbouring lines before Manifold sees the picture, and on interlaced MPEG-2 it blends the fields completely.

Upgrading from 0.8.4

Once you have opened 0.9.0, don't go back to 0.8.4. Your saved streams won't connect in 0.8.4, and editing or deleting one there can lose it.

The first time 0.9.0 opens, it moves a few things by itself. You don't need to do anything.

- **Your license** stays activated; you won't be asked for your key again. Manifold checks it with the license server once, then at most once a week. Your license email and machine ID move into the macOS Keychain and leave the preferences file.
- **Your saved streams** move into the macOS Keychain: the full address and any passphrase. The list still shows each stream's name and server.
- **Your settings** carry over unchanged. New settings, such as Highlight color, start at their defaults.

If macOS won't let Manifold read the Keychain at that moment (for example, the keychain is locked), nothing is lost. Manifold leaves your license and streams exactly as they were, says what happened, and tries again the next time it opens.

Where things moved:

- Export Diagnostics… is in the Help menu.
- Settings now has tabs. The playhead is under Playback, SDI scaling under DeckLink Output, and Pro Video Formats under Sources.
- Several keys changed (see Improved ▸ Interface and shortcuts). Help ▸ Keyboard Shortcuts… shows them all.

For testers: what to look at

- **The upgrade itself.** After upgrading, is your license still active with no prompt to re-enter it? Does every saved stream connect? Did macOS ask for your keychain password at any point? (It shouldn't.)
- **Sync on live streams** over long sessions, on the servers and encoders you really use. If you calibrate, does the measured offset match what you see and hear?
- **SDI audio during file playback.** Earlier builds dropped out every few seconds. Listen through a long file on your card and tell us if you hear any gap other than at a seek, scrub or loop point.
- **HDR and color on a reference monitor over SDI,** especially PQ and HLG sources, and whether your monitor switches modes on its own.
- **Interlaced files** out as 1080i: do fields play in the right order on your monitor?
- **Anything that worked in 0.8.4 and doesn't now.**

Reporting a problem: Help ▸ Export Diagnostics… saves a report we can read (no license details, and no file names unless you tick the box). For a picture problem, Help ▸ Export Frame for Support…

saves the frame itself.

About this tester build: it carries extra diagnostics. There is a Debug menu, but it stays hidden unless we ask you to turn it on: one of its items replaces the program audio with a test tone. A few Control-Option key combinations are test shortcuts and connect on their own: ⌃⌥D and ⌃⌥H connect to your first saved SRT or WHEP stream, and ⌃⌥N to the first NDI source on your network. Please avoid them.